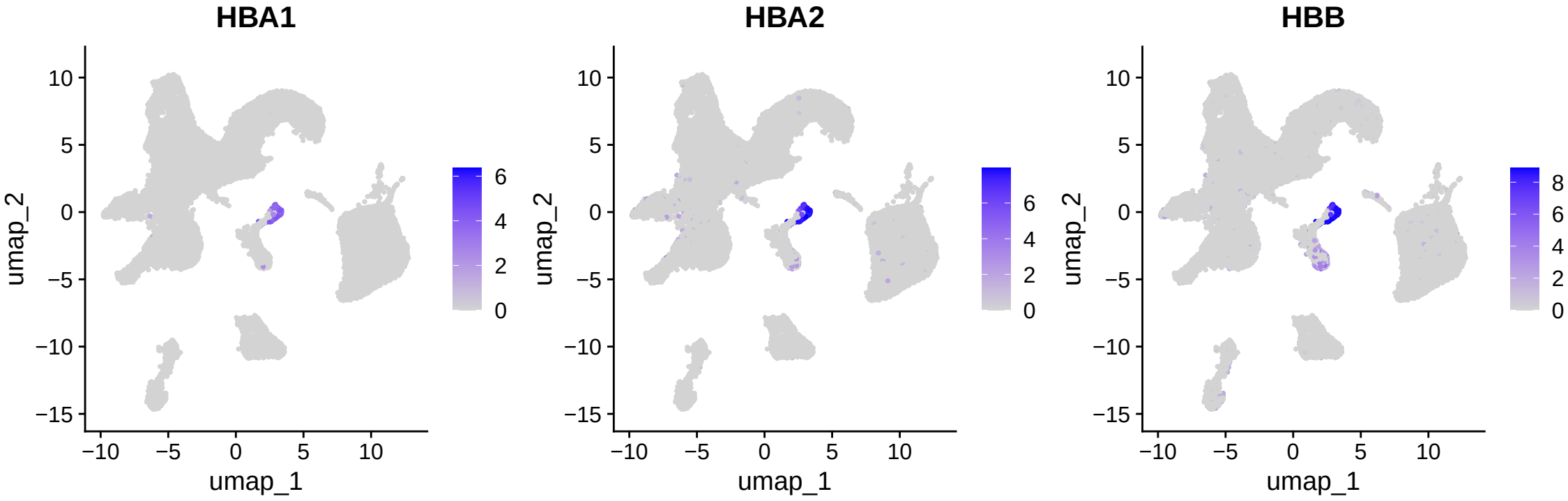

HBA1
HBA2
HBB
umap_1
umap_2
10
5
0
−5
−10
−15
−10
−5
0
5
10
6
4
2
0
8